Impulse Play

Feels like playing? Make it happen.



CSIS-3375-003 UX Design in Web & Mobile App

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1. Introduction

App description

ImpulsePlay is a proposed social sports app represented as an interactive Figma prototype. It is designed to help people find compatible players, automatically form groups, and organize games according to sport, skill level, availability, location, play style, social preference, and group size. Rather than limiting users to games that someone has already organized, the concept begins with individual interest and turns compatible intentions into a proposed game.

The initial sport choices are pickleball, basketball, soccer, tennis, and badminton, which the idea notes identify as choices based on popularity in Canada. A solo player or a pre-made party uses one primary action: Find a Match. The system adds compatible players until the selected participant target is reached. Joining, forming, and filling are system outcomes rather than decisions the user must make.

Problem addressed

- A user may not have enough friends who play the same sport.
- Friends may not be available at the same time.
- An established group may need one or two more players to complete a game.
- Players may struggle to find others at a similar skill level.
- A person may feel uncomfortable joining an established group of strangers.
- Coordinating schedules, venues, and costs can take too much time.

The badminton example in the concept captures the problem: if a group of four loses one player, the entire session may fall apart. ImpulsePlay is intended to help that group fill the missing place or help each interested person become part of a newly formed group.

Novelty

The proposed novelty is intent-based automatic group formation through one matchmaking queue. The project reframes the usual flow—someone creates a game, other people discover it, and people join—into a new flow: a solo player or party expresses interest, the system finds compatible players, and a date and venue are proposed. Its focus is therefore not only discovery, but the conversion of separate intentions into an organized real-world activity.

> **Solo player or party → Find a Match → Confirmation → Payment → Game details**

The matching concept considers sport, skill, overlapping availability, selected cities, play style, social preference, and group size together. Like selecting a matchmaking region in an online game, the party leader can search one or more nearby cities. Friends can enter together as a party, while a player without the app can be represented by a verified temporary guest profile. This reduces the scheduling, social, and coordination effort described in the idea notes. In the prototype, this behaviour will be demonstrated through designed screens and interactions rather than a functioning matching system.

Transformative user experience

The intended transformation happens in three stages:

1. “I don’t have anyone to play with.”
2. “I can play tonight.”
3. “I now have a regular sports community.”

After a game, compatible players can be added to a personal Sports Circle so future games are easier to organize. The longer-term relationship is designed to move from Stranger → Player → Teammate → Regular Sports Community.

Motto

> **Turn “I want to play” into an actual game.**

2. Background Research

Market competitor apps

The idea notes identify Meetup, OpenSports platform, GoodRec, SportsDock, and Sportster as competitors. The comparison below stays within that list. Official pages are used only to confirm the basic discovery or joining pattern; no unsupported features are assigned to Sportster.

Competitor	Relevant existing experience	Relationship to the proposed concept
Meetup	Users discover sports groups and activities. [1]	Represents the group-and-event discovery pattern described in the idea notes.

OpenSports	Users can find open sporting events and create games. [2]	Represents an existing-game and user-created-game flow. This is distinct from the proposed concept despite the shared working name.
GoodRec	Users browse available local games, choose a sport or city, and claim a spot. [3]	Shows a streamlined way to join organized pickup games.
SportsDock	Users review a sport, location, schedule, and available spots before joining. [4]	Shows a join decision based on practical game details.
Sportster	Identified as a competitor in the project idea notes. [5]	Included as named in the notes; no additional product claim is made here.

Opportunity identified by the concept

The proposal's opportunity is the difference between discovering a game and creating one from distributed demand. The proposed experience asks users to state when, where, how, and at what level they want to play. It then demonstrates a compatible group and game proposal. This is the project's intended point of distinction from the competitor pattern described in the idea notes.

Market inspiration apps

The inspiration comes from multiplayer online player-versus-player games such as Dota, League of Legends, Apex, and Valorant. In the pattern described in the idea notes, a player logs in, may create a party with friends, selects matchmaking for an intended region, and is matched with other players at a similar level. ImpulsePlay transfers that perspective to local, real-world sports: a solo player or party selects one or more cities and presses Find a Match, while the system handles whether it must combine players or fill missing positions.

The inspiration is a perspective on interaction flow, not a claim that the proposed Figma prototype will implement a live matchmaking server.

3. PACT Framework

PACT element	Application to ImpulsePlay
People	People who want to play a sport but lack enough available friends; existing groups missing one or more players; solo players looking for compatible participants; people uncomfortable joining an established group of strangers; temporary guests who cannot or do not want to install the app; and players whose preferences range from casual/social exercise to competitive play.
Activities	Enter an availability window during registration; create or join a party; add a temporary guest; select a sport and one or more search cities; choose whether to stop at the minimum or include suggested substitutes; press Find a Match; accept or decline a proposed date and venue; retain accepting players while replacing declined or unpaid positions; pay an equal share; review final game and equipment details; provide lightweight post-game feedback; and manually connect with compatible players through a Sports Circle.
Contexts	Local sports participation, including the Richmond/Vancouver example in the idea notes; solo or pre-made-party matchmaking; situations where friends still need additional players; cases where casual players want substitutes available when someone becomes tired; coordination and payment before a game; and brief feedback after a game.
Technologies	Figma is the project technology. The prototype will visually simulate party creation, temporary guest profiles, multi-city matchmaking, overlapping availability, acceptance and replacement states, sport-based participant suggestions, venue selection, split payments, equipment information, team balancing, and game-status transitions. It will not be a coded or operational app.

PACT implications for the design

- The flow should reduce the amount of organizing a user must do before seeing a playable option.
- Skill, play style, and social preference should be visible enough to make a proposed group feel compatible.
- Find a Match should be the single primary action; joining, forming, and filling should remain invisible system outcomes.

- Party members with accounts should confirm themselves, while the leader confirms temporary guests and supplies their information.
- The proposed game should make the date, time, nearby facility, group size, compatibility, and estimated per-person cost easy to review.
- Players who accept should remain in the same temporary party while declined or unpaid positions are refilled.
- Post-game feedback should stay lightweight so that attendance, skill accuracy, playing style, and willingness to play again can gradually inform player profiles.
- The Sports Circle should connect a successful first game to easier future organization and a regular sports community.

4. Value Proposition

> **For solo players and groups that want to play a local sport but still need compatible participants, ImpulsePlay provides one matchmaking action that turns shared availability and preferences into a confirmed, paid game. By matching sport, skill, schedule, selected cities, play style, social preference, and group size—and by retaining accepted players while replacing missing ones—it reduces coordination, payment, and social barriers.**

The value comes from combining immediate utility with accountability and the possibility of longer-term connection. Immediate utility is the movement from wanting to play to receiving a workable date and venue, paying an equal share, and receiving complete match information. Reliability consequences discourage repeated declines and objective guest failures. After the game, lightweight feedback and meaningful profile labels remain available, while players decide for themselves whether to connect through the Sports Circle.

5. Project Contract

Purpose. This agreement sets out how our team will work together to design the ImpulsePlay Figma prototype, so that responsibilities, communication and expectations are clear to everyone.

Team members and roles

Member	Student Id	Main responsibilities
Steven Anderson	300414292	Project Proposal, Research, Implement

Hung Le	300417350	Project Proposal, Research, Implement

Everyone contributes to report writing and review. Roles may shift if workload changes, with the agreement of the team.

Communication. We will use Discord or Whatsapp for daily communication and meet weekly once a week on Tuesday. Members will reply to team messages within 12 hours.

Figma file management. Hung owns the main Figma file. Members work in their own pages or a copy and merge into the main file after a quick check with the team.

Decision making. We aim for consensus. If we disagree, we vote by majority; if still unresolved, we ask the instructor.

Deadlines and work logs. Each member records dates, hours and tasks in the Work Log (Section 9) after each work session. Tasks are due on the dates agreed in the team timeline.

Accountability. Members who cannot attend or finish a task will tell the team as early as possible. Repeated missed work will first be discussed within the team and, if not resolved, reported to the instructor.

Use of AI tools. AI tools may be used for [e.g., organizing notes, improving wording], and every use is recorded in Section 8 and Appendix A. Design decisions and final content remain the team's own.

Agreement. By signing below, each member agrees to these terms.

Name	Signature	Date
Steven Anderson		10/9/2026
Hung Le		10/9/2026

6. Preliminary Feature List

This list is preliminary and may evolve during the design process. Every feature below comes from the supplied idea notes and is represented only at prototype level in Figma.

Sport selection: Pickleball, basketball, soccer, tennis, and badminton as the initial choices.

Required availability: Registered users enter an availability window before they can use matchmaking.

Party creation: A player can enter alone or invite friends and search as a party.

Temporary guest profiles: A leader can add a person who does not have the app through a lightweight profile verified by phone or email. The leader confirms the guest, and the guest uses a browser link for payment and match details.

One Find a Match action: The user does not choose whether to join, form, or fill a game. The system treats those as internal matching outcomes.

Multi-city search: The party leader selects one or more acceptable cities before searching, similar to choosing matchmaking regions.

Intent entry: Sport, availability, skill, selected cities, play style, social preference, and group size.

Intelligent matching: A simulated match based on the entered criteria and the overlap between participant availability windows.

Dynamic group formation: A visual flow from solo players and pre-made parties to a proposed playable group.

Participant target setting: Before searching, the party chooses Minimum Players Only or Include Suggested Substitutes. A sport-based suggestion determines a reasonable target, and the choice cannot change during the search.

Game proposal: A proposed date, time, suitable nearby facility, compatible player group, and estimated cost per person.

Accept or decline: Registered players decide whether to accept the proposal. Repeated declines of suitable proposals can reduce reliability, apply a temporary cooldown, and eventually require email reverification.

Temporary matched party: Players who accept stay together at the proposed date and location while the system replaces anyone who declined.

Replacement matching: The system refills positions left by a declined response or missed payment without restarting accepted players.

Suggested substitutes: When selected before the search, matchmaking continues beyond the minimum to provide a reasonable number of participants who can substitute during casual play.

Interactive player compatibility: Prototype interactions that help users understand why players fit the proposed game.

Venue selection: A prototype choice or recommendation for a suitable nearby facility.

Equal bill splitting: The venue cost is divided equally among every participant, including substitutes.

Individual payment: Each participant receives a payment request. The system pays the venue after every share has been covered.

Guest payment responsibility: A temporary guest can pay through the browser link. If the guest does not pay by the deadline, the inviting leader must cover the share or the guest is replaced and the objective failure affects the leader's guest-reliability record.

Final match information: After successful payment, players receive the date, time, venue, participant, payment, and equipment details, including whether they must bring equipment or can borrow it at the facility.

Team balancing: A prototype representation of balanced teams.

Game-status transitions: Prototype states that show a game moving through its relevant stages.

Lightweight post-game feedback: Attendance, skill-level accuracy, casual-to-competitive style, and willingness to play together again.

Meaningful player labels: A developing profile that can show level, attendance percentage, play-style label, and games played.

Guest accountability: Objective guest failures, such as a confirmed no-show or uncovered payment, affect the inviting leader; subjective skill and play-style ratings remain attached only to the guest.

Sports Circle: The temporary party dissolves after the game. Players who feel a connection can manually add one another to a Sports Circle for future games.

7. Figma Prototype Scope

The project deliverable is an interactive Figma prototype only. Its purpose is to communicate the intended user journey, information, matching logic, and state changes through designed screens and clickable interactions. It does not include a coded application or an operational matching service.

The prototype will demonstrate:

- registering with an availability window and updating matchmaking preferences;
- entering alone or creating a party with registered friends and temporary guests;
- selecting one or more cities and choosing Minimum Players Only or Include Suggested Substitutes before the search;
- pressing a single Find a Match button;
- seeing a proposed date, venue, compatible participant group, and estimated individual cost;
- accepting or declining and retaining accepted players while declined positions are refilled;
- showing reliability warnings, cooldown, and email-reverification states without assigning invented numeric thresholds;
- continuing to a sport-based participant target when substitutes were enabled before matchmaking;

- collecting equal payment shares, handling registered-player and temporary-guest payment failures, and showing automatic venue payment after all shares are covered;
- reviewing confirmed match and equipment details;
- providing the four lightweight types of post-game feedback; and
- dissolving the temporary party while allowing players to connect manually through the Sports Circle.

Illustrative solo scenario

User intent	Proposed result
Badminton; tonight after 7:00 p.m.; intermediate; Richmond/Vancouver; social + exercise.	Game found; four compatible players; 7:30–9:30 p.m.; suitable badminton facility nearby; similar skill and play style; estimated cost of $9 per person.

Illustrative party scenario

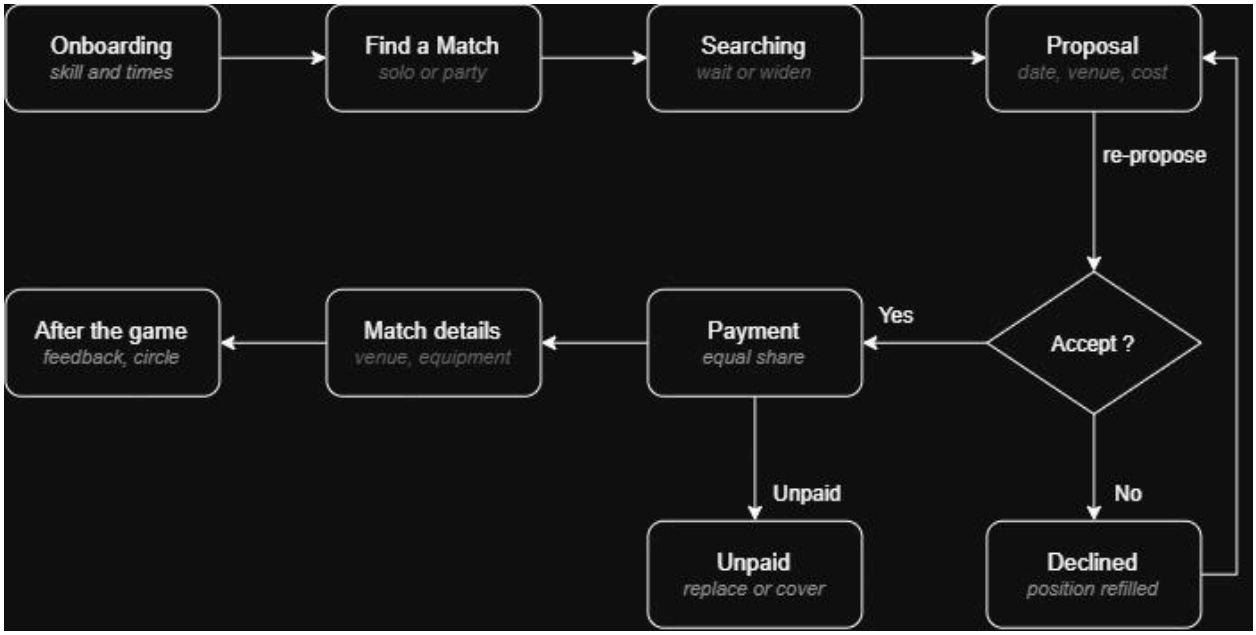
Stage	Prototype behaviour
Party setup	A leader invites registered friends and may add a verified temporary guest. Registered members confirm themselves; the leader confirms the guest, who shares the leader's availability.
Search setup	The leader selects one or more cities and chooses Minimum Players Only or Include Suggested Substitutes before pressing Find a Match.
Proposal	The system finds compatible players and proposes one date and venue from the overlapping availability and selected cities.
Decision	Accepting players remain in a temporary party. Declining players leave, receive the relevant reliability consequence when repeated, and are replaced.

Participant completion	The system stops at the minimum or continues to the sport-based suggested participant target, according to the setting chosen before the search.
Payment	The cost is split equally. Unpaid registered players are replaced. If a guest does not pay, the leader receives a deadline to cover the guest's share before replacement and a guest-reliability consequence.
Confirmation	After every share is covered, the prototype shows venue payment and complete match information, including equipment requirements or borrowing availability.
After the game	The temporary party dissolves. Players decide whether to connect manually through the Sports Circle.

Core user flows and screen scope

The prototype follows one main journey from onboarding to a confirmed game, plus the side paths where a player declines or does not pay.

One Find a Match action leads to a paid, confirmed game



Edge Cases

Situation	Planned design response
Searching, with no match yet	A waiting screen with estimated progress and the option to be notified when a match is found
No match found	A message with suggestions to widen cities, time window or skill range
Proposal not answered	A visible countdown on the proposal; expiry handling counts as a decline.
Player declines or leaves	Accepted players stay in the temporary party while the system refills the position
Substitutes added	Updated cost per person shown before any payment request [to decide: when the participant target is final]

Payment not completed	Deadline and reminder; unpaid registered player is replaced; guest share is covered by the leader or the guest is replaced
Player drops out after paying	Clear refund or credit rule shown at payment [to decide]
Venue cancels or weather affects an outdoor game	Notification with reschedule or refund options [to decide]

Returning users. After the first game, a returning player should be able to start a new search from saved preferences, so the single Find a Match action stays quick on repeat use.

8. AI Use

Table of AI tools and specific use

Tool name, version, and account type	Specific use
OpenAI Codex (GPT-5.6 - Sol); account type: premium	Used to organize the supplied Project Requirement and ImpulsePlay Idea notes and the student's follow-up design decisions into proposal sections; improve wording and grammar; generate images; format an editable Word document; and locate official pages for the competitor apps already named in the idea notes. The concept and feature ideas remain those supplied by the student/team.
Claude (Sonnet 5.5); Account type: Pro	Used to fine tune the scope, improve wording and grammar. Cross checking with other AI tools. The concept and feature ideas remain those supplied by the student/team.

The account type is left as a placeholder because it was not provided. Replace the bracketed text before submission.

9. Work Date/Hours Logs

Complete one work log for each student or team member. Duplicate the table as needed.

Student / team member: Steven Anderson

Date	Start time	End time	Hours	Work completed
8/10/26	3:00 AM	5:15 AM	2 H 15 M	Project Proposal + Revision
9/10/26	9:00 AM	2:30 PM	2 H 30 M	Completing Document
[insert]	[insert]	[insert]	[insert]	[insert]
[insert]	[insert]	[insert]	[insert]	[insert]
[insert]	[insert]	[insert]	[insert]	[insert]

Total hours: [insert]

Student/ team member: Hung Le

Date	Start time	End time	Hours	Work completed
8/10/26	11:00 PM	1:00 AM (9/10/26)	2 H	Project Proposal + Revision
[insert]	[insert]	[insert]	[insert]	[insert]
[insert]	[insert]	[insert]	[insert]	[insert]
[insert]	[insert]	[insert]	[insert]	[insert]
[insert]	[insert]	[insert]	[insert]	[insert]

Total hours: 2 hours

10. Closing and References

Closing

ImpulsePlay proposes a Figma experience that moves beyond searching for a pre-existing activity. It gives a solo player or pre-made party one Find a Match action, then demonstrates how compatible participants can reach an accepted date and venue, maintain the committed group while filling missing positions, divide the cost, and receive complete match information. Its central design purpose is to reduce the matching, scheduling, payment, and social barriers that prevent people from playing.

Acknowledgements

[Insert acknowledgement if needed.]

References

[1] Meetup. “Sports and Recreation.” https://www.meetup.com/topics/sports/ (accessed October 8, 2026).

[2] OpenSports. “OpenSports – Pickup Sports.” Google Play. https://play.google.com/store/apps/details?id=opensports.android (accessed October 8, 2026).

[3] GoodRec. “Pickup Sports Games Near You.” https://www.goodrec.com/ (accessed October 8, 2026).

[4] SportsDock. “SportsDock Web App.” https://sportsdock.ca/en/web-app (accessed October 8, 2026).

[5] ImpulsePlay Idea.txt. Unpublished project idea notes supplied by the student/team, October 8, 2026.

[6] Project Requirement.txt. Project proposal requirements supplied by the student/team, October 8, 2026.

Appendix A: AI Prompt History

The following task prompts were provided to the AI tool during preparation of this document.

Prompt 1

> help me make a project proposal words with the requirement from project requirement.txt, all my ideas is in ImpulsePlay Idea.txt. Don't invent something that is not in the ideas, let me know if you have any question. Leave the project contract section blank, I will edit that part. Put the title page as placeholder also, just put the app name.The project only going to be made in Figma as prototype

Prompt 2

> why you put 3 paths for user to choose rather than just finding match like how online pvp game works like dota?

Prompt 3

> so if people have friends to play but missing more people they can just create a party then find a match together. they will automatically find a match nearby their location/ city but they can change or add more city to find. like how dota player can change their region for matching. then after they found enough people, the system will propose the date and location for the play and the players will decide whether to accept or decline. declining too much will punish them. if for example everyone accept but one decline, they will be pooled in the same "temporary party" to find the missing player. after they put into the date and location that have been chosen, the system will keep finding more people to participate until a reasonable amount of participant is reached for substitution, since people who lean into casual play tend to substitute player when they're tired to maximize the usage of the field. we can also make an option to not find more people if minimum player reached. Then after all of that, the system will split the bill for every participant. Like how online order works, it will ask each participant how to pay the bills. After all participant paid successfully, the system will pay it automatically and give the details information to the players regarding the match, also if its something like badminton where you need racket and shuttle cock the details will tell the user to bring the equipment themself or if the place provide borrowing service for the equipment. That's what I'm thinking.

Decision response 1

> Decline consequence: both a temporary cooldown and a reliability score, with email reverification after excessive declines. Search-area control: the party leader chooses the city or cities. Payment failure: replace the unpaid player. Additional idea: a leader can invite someone without the app through a temporary profile, with accountability for bringing unreliable guests.

Decision response 2

> Temporary guest participation: browser invitation link. Guest identity: verified phone or email. Extra-player target: sport-based suggestion, but the party must choose the option before matchmaking and cannot stop it after matchmaking begins.

Decision response 3

> Party readiness: registered members confirm themselves; temporary guests require a different rule because they do not have an account. Availability: every registered user must enter an availability window before matchmaking, and the system uses overlapping schedules. Temporary party: dissolve it after the game and let players form longer-term groups manually if they connect.

Decision response 4

> Temporary guests are confirmed by the party leader and are assumed to have the same availability as the leader; registered members confirm themselves. Only objective guest failures affect the leader. If a guest fails to pay, the leader must cover the remaining share before a deadline.

Prompt 6

> Implement the plan.

Prompt 7

> I just add new flow user chart, contract, and could you make an edges case where searching, no match found/ yet where message and waiting screen handle. When the proposal not answered which we should put a countdown or expiry handling. When player declines or leaves, substitutes added which updated cost when the target is final. When payment not completed, reminder will pop up and replace the player. Refund when drop out or cancel when weather affects. And if possible please add more safety, communication (still deciding) and repeated declines.

Decision response 7

> The prototype includes the following situations beyond the main path:
>
> 1. Searching, and no match found or not yet. A waiting screen shows search progress and offers a notification when a match is found. If nothing is found, a message suggests widening cities, time or skill range.
> 2. Proposal not answered. Each proposal shows a countdown. When it expires, the proposal is handled as unanswered and the position is refilled.
> 3. Player declines or leaves. Accepted players stay in the temporary party while the system refills the position.
> 4. Substitutes added. The cost per person updates when the participant target is final, and players see the new cost before paying.
> 5. Payment not completed. A reminder appears before the deadline; if the share is still unpaid, the player is replaced (for a guest, the leader may cover the share).
> 6. Dropout, cancellation or weather. Players see clear refund rules when they drop out or when the game is cancelled because of weather.
> 7. Safety. Report and block actions, profile verification indicators, and age and waiver consent screens.
> 8. Communication. A group message or notification thread after confirmation; the choice between chat and notifications is still being decided.

9. **Repeated declines. Warning, cooldown and email-reverification states, without numeric thresholds in the prototype.**